

EE 156: FUNDAMENTALS OF ELECTRICAL CONTROLS AND PLC SYSTEMS

Chapter 7: Programming Logic Controller (PLC)

7.1 Introduction

What is Industrial Automation?

- Use of control systems such as:
 - Numerical control
 - Programmable logic control
 - Other industrial control systems
- Combined with information technology (CAD, etc.)
- Controls industrial machinery and processes
- **Reduces need for human intervention**
- Step **beyond mechanization**

Why PLCs in Automation?

- **Specialized hardened computers** used to synchronize:
 - Inputs from sensors and events
 - Outputs to actuators and events
- Permits **precisely controlled actions**
- Allows **tight control** of almost any industrial process

Human-Machine Interface (HMI)

- Also called Computer Human Interface (CHI)
- Used to **communicate with PLCs**
- Enter and monitor temperatures, pressures
- Allows for further automated control or emergency response
- Personnel who monitor these interfaces are called **stationary engineers**

7.2 Programmable Logic Controller (PLC)

Definition

A PLC is an **industrial grade computer** that is capable of being programmed to perform control functions.

Key Characteristics

Feature	Description
Industrial grade	Designed for harsh environments
Programmable	Can be reprogrammed for different tasks
Real-time system	Output depends on input conditions

Feature	Description
Multiple I/O	Designed for many inputs and outputs

Why PLCs are Real-Time Systems

- The **output** depends on the **input conditions**
- Must respond within a **defined time frame**
- Continuous monitoring and control

7.3 Parts of a PLC

Major Components

Component	Function
Programming Device	Used to enter the program
Power Supply	Provides DC power to modules
I/O Section	Interface to field devices
CPU (Processor)	"Brain" of the PLC

Architecture Types

Type	Description
Open Architecture	Can be modified/expanded
Closed Architecture	Fixed configuration

7.4 Input/Output (I/O) Modules

Purpose of I/O System

- Forms the **interface** by which control devices are connected to the controller
- Allows PLC to:
 - **Receive inputs** from sensors and other devices
 - **Provide outputs** to actuators and external devices
- Acts as a **bridge** between:
 - Digital/analog signals from field devices
 - Internal processing unit of the PLC

I/O Configuration Types

Type	Description	Application
Fixed I/O	I/O is built-in, cannot be expanded	Small, simple systems
Modular I/O	I/O modules can be added/removed	Large, flexible systems

Input Modules

Function: Receive signals from field devices

Examples of Input Devices:

Device	Type	Function
Sensors	Digital/Analog	Detect conditions
Switches	Digital	Manual input
Push buttons	Digital	Operator commands
Encoders	Digital	Position/speed sensing
Thermocouples	Analog	Temperature sensing

Output Modules

Function: Send control signals from PLC to external devices

Examples of Output Devices:

Device	Function
Motors	Motion control
Valves	Fluid control
Solenoids	Actuation
Lights	Status indication
Contactors	Power switching

7.5 PLC Power Supply

Function

- Supplies **DC power** to modules that plug into the rack
- Rectifies input AC voltage to DC
- Filters and regulates voltage

Typical Voltage Levels

- Internal logic: **5 to 24 volts DC**
- Field devices: May use **external AC or DC supplies**

Power Supply Role

AC Input -> Rectification -> Filtering -> Regulation -> DC Output

Larger Systems

- Power to field devices provided by:
 - External AC supplies
 - External DC supplies
- Separate from PLC power supply

7.6 PLC Processor (CPU)

Function

- The **"brain"** of the PLC
- Consists of a **microprocessor** for:
 - Implementing logic
 - Controlling communications among modules

CPU Control Activities

Activity	Description
Program execution	Runs user program
I/O control	Reads inputs, updates outputs
Communications	Interacts with other devices
Memory management	Stores program and data
Diagnostics	Self-checking and fault detection

Memory Requirements

- User program instructions
- Numerical values
- I/O device status

The Scan Cycle

Definition: The repetitive process by which the PLC executes the program

Scan Cycle Steps

1. Read Inputs
 2. Execute Program
 3. Update Outputs
 4. Housekeeping/Communications
- (Repeat)

Step	Action
1. Read Inputs	CPU reads status of input modules
2. Execute Program	CPU processes user program logic
3. Update Outputs	CPU writes results to output modules
4. Housekeeping	Communications, diagnostics

7.7 Programming Device

Function

- Used to **enter the desired program** into the memory of the processor

Most Common Programming Device

- **Personal Computer (PC)**
- Communicates with PLC processor via:
 - Serial data communications link
 - Parallel data communications link
 - Ethernet

Program Definition

- A **user-developed series of instructions**
- Directs the PLC to **execute actions**

7.8 PLC Programming Languages

Overview

- Method by which the user **communicates information** to the PLC
- Allows programmers to **define the behavior** of the PLC
- Designed to be:
 - User-friendly
 - Efficient
 - Capable of handling real-time control tasks

IEC 61131 Programming Languages

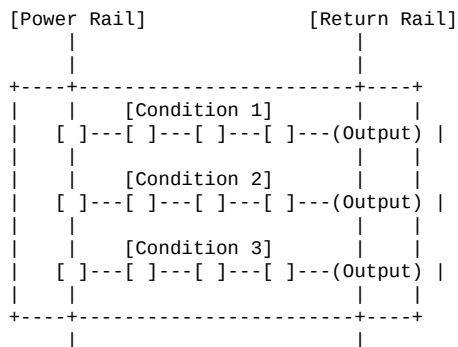
Language	Abbreviation	Description
Ladder Logic	LD	Graphical, resembles relay diagrams
Function Block Diagram	FBD	Graphical, block-based
Structured Text	ST	Text-based, high-level language
Instruction List	IL	Text-based, low-level
Sequential Function Chart	SFC	Graphical, step-based

Ladder Logic (LAD/LD)

Overview:

- Most common PLC programming language
- Uses components that resemble **line diagram format**
- Describes **hard-wired control**

Ladder Diagram Structure



Ladder Logic Components

Symbol	Component	Description
-[]-	Normally Open (NO)	Closed when condition true
-[/]-	Normally Closed (NC)	Open when condition true
-()-	Output Coil	Energized when rung true
-()-	Output	Energized when rung true

Ladder Logic Rules

Rule	Description
1	Read from left-to-right, top-to-bottom
2	Left rail = Power/Energized conductor
3	Right rail (return path) is omitted
4	Rungs are sometimes called networks
5	Network may have multiple control elements
6	Network has only one output coil

7.9 Example of PLC Ladder Connection

Step 1: Hard-Wired Circuit

Circuit Description:

- Pushbutton controls a motor starter
- Motor starter controls a motor

Hard-Wired Diagram:

L1 -----[PB]-----[M]----- N

Where:

PB = Pushbutton (NO)

M = Motor starter coil

Step 2: PLC Ladder Equivalent

Ladder Logic:

L1 -----[PB]-----[M]----- N

PB = Input (I:0/0)

M = Output (O:0/0)

Step 3: PLC Addresses

Component	Symbol	PLC Address
Pushbutton	PB	I:0/0 (Input 0, bit 0)
Motor Starter	M	O:0/0 (Output 0, bit 0)

Step 4: Example with Multiple Inputs

Start-Stop Circuit:

L1 -----[Stop]-----[Start]-----[M]----- N
 | | |
 +-----[M]---+ |

Where:

Stop = NC pushbutton

Start = NO pushbutton

M = Motor starter (with seal contact)

PLC Equivalent:

I:0/0 I:0/1 O:0/0
--[Stop]--[Start]----- (Motor) --
 | |
 +-----[Motor]-----+

7.10 PLC I/O Addressing

Input Addressing

PLC Type	Example Address	Description
Allen-Bradley	I:0/0	Input module 0, bit 0
Siemens	I0.0	Input byte 0, bit 0
Modicon	00001	Input 1

Output Addressing

PLC Type	Example Address	Description
Allen-Bradley	O:0/0	Output module 0, bit 0
Siemens	Q0.0	Output byte 0, bit 0
Modicon	00001	Output 1

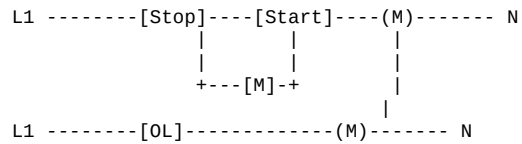
7.11 PLC Programming Example

Example: Motor Starter Control

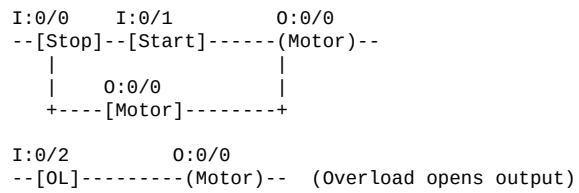
Requirement:

- Start pushbutton starts motor
- Stop pushbutton stops motor
- Motor stays running after start button released (seal circuit)
- Overload protection

Hard-Wired Ladder



PLC Ladder



7.12 PLC vs Hard-Wired Control

Feature	Hard-Wired Control	PLC Control
Flexibility	Fixed, changes require rewiring	Easily reprogrammed
Troubleshooting	More difficult	Easier (diagnostic tools)
Space	Requires large panels	Compact
Cost	Lower for simple tasks	Higher initially
Maintenance	More wiring to check	Less wiring
Reliability	Good	Excellent
Documentation	Manual	Automatic and online
Scalability	Difficult	Easy

7.13 Key Terminology

Term	Definition
PLC	Programmable Logic Controller
CPU	Central Processing Unit (processor)
I/O	Input/Output
Ladder Logic	Graphical PLC programming language
Scan Cycle	Repetitive process of reading inputs, executing program, updating outputs
Rung	Horizontal line in ladder diagram
Network	Another name for a rung
Seal Circuit	Circuit that maintains itself
HMI	Human-Machine Interface
Real-time System	System where output depends on input conditions

Review Questions

1. What is a PLC and why is it used in industrial automation?
2. List and explain the major parts of a PLC system.
3. What is the function of the I/O modules in a PLC system?
4. Explain the PLC scan cycle.
5. What is ladder logic and why is it commonly used?
6. How does a PLC ladder diagram compare to a hard-wired relay diagram?
7. What is the purpose of a seal circuit in ladder logic?
8. List the IEC 61131 programming languages.
9. What is the function of the CPU in a PLC?
10. What is the difference between fixed and modular I/O?